FORE SCHOOL OF MANAGEMENT

A Technology Stack Guide

on

“ShopAssist-AI: An Agentic RAG Customer-Support and Inventory Assistant for an Online Clothing Store”

(Every tool, model, method and n8n node used in the project – what it is, why it was chosen and how it is used)



Course: Agentic AI for Business Automation

Capstone Project (Individual)

Submitted by

065087: Kritika Bhachawat

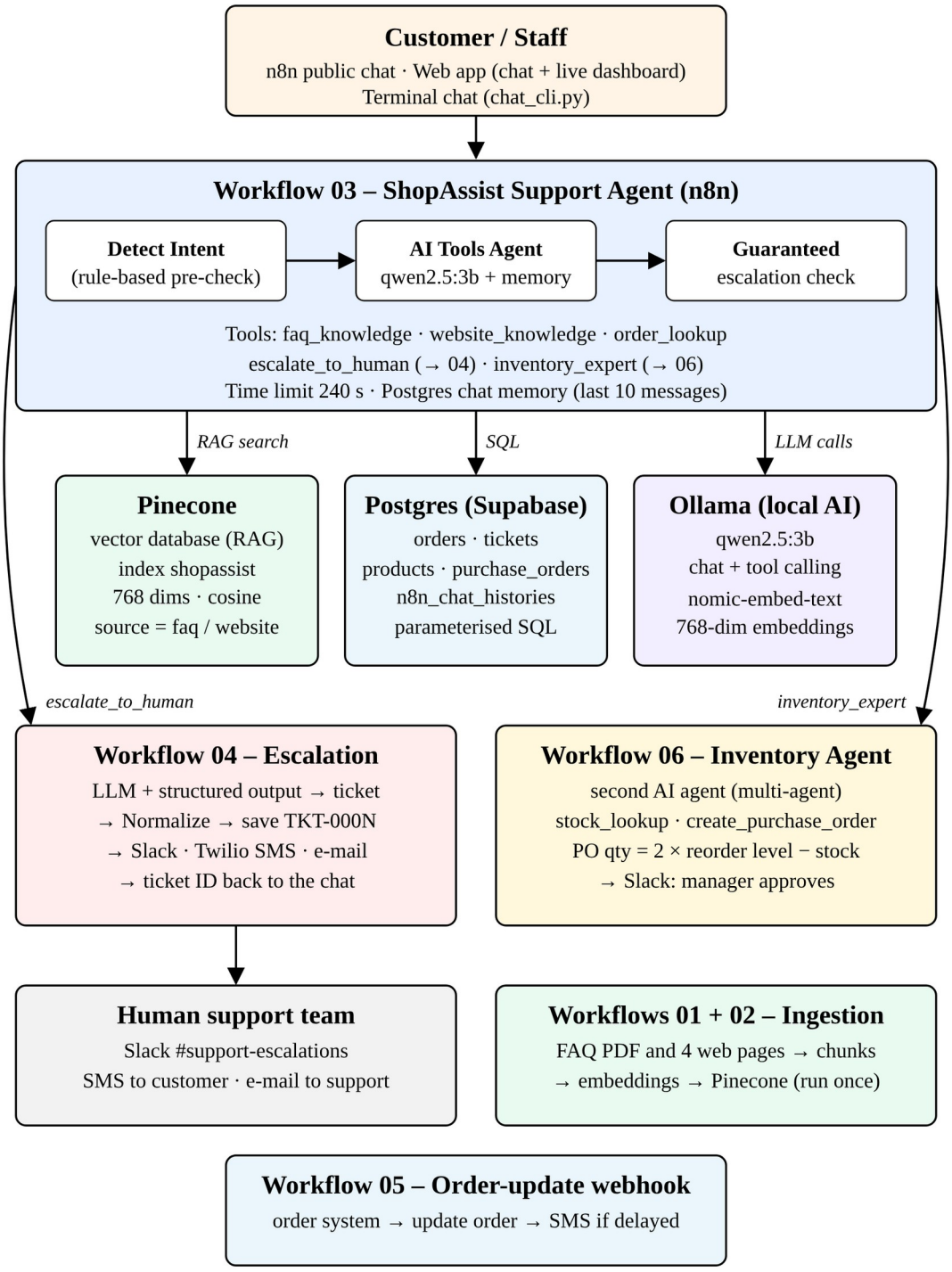
PGDM (Business Data Analytics), 2025–27

FORE School of Management, New Delhi

Date of Submission: 10th October 2026

Submitted to: Prof. Shiva Botchu

1. The stack at a glance



How the technologies fit together

Layer	Technology	Cost	Runs where
Automation & agents	n8n 2.x	Free (self-hosted)	Laptop, Docker
Language model	qwen2.5:3b via Ollama	Free	Laptop
Embedding model	nomic-embed-text via Ollama	Free	Laptop
Vector database	Pinecone	Free tier	Cloud

Layer	Technology	Cost	Runs where
Relational database	PostgreSQL (Supabase)	Free tier	Cloud
Notifications	Slack, Twilio SMS, Gmail SMTP	Free / trial	Cloud
App & tooling	Python 3, HTML/CSS/JS, Docker, Git/GitHub	Free	Laptop

2. Core AI concepts used

2.1 Agentic AI and AI agents

Aspect	Details
What it is	An **AI agent** is an LLM that can decide which actions (tools) to take, observe the results and continue until the task is done, instead of only producing text.
Method	Reason → act (call a tool) → observe → answer. This is the **ReAct** pattern; n8n's Tools Agent implements it with native tool calling.
In ShopAssist	Two agents: the **Support Agent** (workflow 03, 5 tools, up to 8 steps) and the **Inventory Agent** (workflow 06, 2 tools, up to 4 steps).
Why	Customer questions vary; the agent chooses between the FAQ, website, order database, inventory expert and escalation itself.

2.2 Tool calling

Aspect	Details
What it is	The model returns a structured request such as order_lookup({"order_id_or_phone":"UT1002"}) instead of text. n8n runs the tool and gives the result back to the model.
In ShopAssist	Tools: faq_knowledge, website_knowledge, order_lookup, escalate_to_human, inventory_expert, stock_lookup, create_purchase_order.
How inputs are filled	n8n $fromAI(name, description, type, default) lets the model fill each tool parameter. Defaults keep a tool working if the model forgets a field.
Why it matters	The model needs **tool-calling support**, which is why qwen2.5 was chosen.

2.3 RAG – Retrieval-Augmented Generation

Aspect	Details
What it is	Before answering, search a knowledge base for the most relevant text and give it to the LLM, so the answer is based on the company's own documents, not on the model's memory.
Steps	1) Load documents → 2) split into chunks → 3) turn chunks into vectors (embeddings) → 4) store in a vector database → 5) at question time embed the question, find the nearest chunks, answer from them.
In ShopAssist	Two knowledge bases in one Pinecone index, separated by metadata: source=faq (PDF) and source=website (4 scraped and summarized pages).
Agentic RAG	The agent decides **which** knowledge base to search (or none), instead of always searching everything.
Benefit	Fewer invented answers ("hallucinations"), easy updates (re-ingest a document, no retraining), and answers can name their source.

2.4 Embeddings and vector search

Aspect	Details
What it is	An embedding is a list of numbers (here **768**) that represents the meaning of a text. Texts with similar meaning have vectors that point in similar directions.

Aspect	Details
Similarity	**Cosine similarity** measures the angle between two vectors (1 = same meaning). The Pinecone index uses the cosine metric.
Rule	The same embedding model must be used for storing and for searching, and the index dimension must match the model (768 for nomic-embed-text).
topK	The agent receives the 4 most similar chunks per search (topK = 4).

2.5 Chunking (text splitting)

Aspect	Details
What it is	Long documents are cut into smaller pieces so each vector represents one topic and fits the model's context.
Method	**Recursive Character Text Splitter**: it tries to split at paragraphs, then sentences, then words, so pieces stay readable.
Settings	Chunk size **1,000** characters, overlap **200**: the overlap keeps a sentence that crosses a border searchable in both chunks.

2.6 Metadata filtering

Aspect	Details
What it is	Each vector carries labels (metadata). A search can be limited to vectors with certain labels.
In ShopAssist	source (faq / website), plus url and title for web pages. faq_knowledge filters source=faq; website_knowledge filters source=website.

2.7 Summarization before storing

Aspect	Details
What it is	Web pages are summarized by the LLM into factual bullets before they are embedded.
Why	Raw HTML text has menus and repetition. Summaries are shorter and cleaner, so retrieval is more precise and uses less context.
Settings	Basic LLM Chain, qwen2.5:3b, temperature 0 (factual, repeatable).

2.8 Structured output

Aspect	Details
What it is	Forcing the LLM to answer in a fixed JSON format defined by a **JSON schema**.
In ShopAssist	Workflow 04 turns an escalation into a ticket with fixed fields: customer_name, phone, order_id, issue_type, sentiment, priority, summary, suggested_action.
Methods	Structured Output Parser (schema) + Ollama JSON mode (format: json) + a **Normalize** step that fills gaps and clamps values to the allowed lists.

2.9 Conversation memory

Aspect	Details
What it is	The agent remembers earlier messages in the same chat session.

Aspect	Details
In ShopAssist	**Postgres Chat Memory**: table n8n_chat_histories, last **10** messages per session, so "and when will it arrive?" works after an order ID was given.

2.10 Multi-agent system

Aspect	Details
What it is	Several specialised agents work together; one agent calls another like a tool.
In ShopAssist	The Support Agent calls the **Inventory Agent** through the inventory_expert tool (n8n "Call n8n Workflow" tool). Each agent has its own prompt, tools and limits.
Why	Separation of concerns: customer service and procurement need different knowledge and permissions. Each agent stays small and reliable.

2.11 Human-in-the-loop

Aspect	Details
What it is	The AI hands over to a person for decisions it should not make alone.
In ShopAssist	Escalation tickets for angry customers and refunds above Rs 5,000; purchase orders are drafts that a manager approves in Slack.

2.12 Guardrails (deterministic rules around the AI)

Aspect	Details
What it is	Rules in normal code that check or correct what the AI does.
In ShopAssist	Detect Intent (pre-check notes and flags), guaranteed escalation, refund amount read from the database, Normalize Ticket, SQL-enforced PO quantity, tool input defaults, fallback replies, time limits, parameterised SQL.
Result	Evaluation went from 9 / 14 to **14 / 14**.

2.13 Prompt engineering

Aspect	Details
What it is	Writing clear instructions (system prompt) that define the agent's role, rules and output style.
In ShopAssist	prompts/agent_system_prompt.md: answer only from tools, cite the source, ask for the order ID, escalation rules, refusal of off-topic questions, one-word message mapping. prompts/inventory_agent_prompt.md: stock and PO rules.
Lesson	Short, explicit notes added to the message (by Detect Intent) worked better for a 3B model than a longer system prompt.

3. Platforms and models

3.1 n8n (workflow automation platform)

Aspect	Details
What it is	An open-source, low-code workflow automation tool. Workflows are built from nodes on a canvas; it has native AI (LangChain-based) nodes.
Version / setup	n8n 2.x, self-hosted in **Docker** (container shopassist-n8n, http://localhost:5679, WEBHOOK_URL set so chat webhooks use the right port).
In ShopAssist	6 workflows: 01 ingest PDF, 02 scrape + summarize, 03 support agent, 04 escalation, 05 order webhook, 06 inventory agent.
Why	Visual and explainable, many integrations (Slack, Twilio, Postgres, Pinecone), webhooks and a public chat built in; free when self-hosted.
Features used	Publishing (activation), sub-workflows, execution log, per-workflow time limits, credentials store, public REST API (used by the import and doctor scripts).

3.2 Ollama (local model runtime)

Aspect	Details
What it is	A free program that downloads and runs open-source LLMs on your own computer and serves them on http://localhost:11434.
Why chosen	Google Gemini returned "403 project denied" for the available keys. Ollama needs no key, costs nothing and keeps customer data on the laptop.
Connection	n8n credential "Ollama account": http://localhost:11434, or http://host.docker.internal:11434 from Docker.

3.3 qwen2.5:3b (chat / reasoning model)

Aspect	Details
What it is	An open-weights LLM from Alibaba's Qwen team, 3 billion parameters (about 2 GB), with native tool calling and JSON output.
Used for	Both agents' reasoning and answers, page summaries (02), ticket JSON (04).
Key settings	temperature 0.2 for the support agent, 0 for the Inventory Agent, summaries and tickets; **numCtx 8192** (the default 2048 cut off the agent prompt); **think: false**; format: json for tickets.
Trade-off	Small enough for a laptop, but it can skip steps, so the guardrails were added. qwen2.5:7b or 14b would be more accurate but slower.

3.4 nomic-embed-text (embedding model)

Aspect	Details
What it is	An open embedding model from Nomic AI, served by Ollama; it produces **768-dimension** vectors and supports long inputs.
Used for	Embedding the FAQ chunks, page summaries and every search question.

3.5 Pinecone (vector database)

Aspect	Details
What it is	A managed cloud database for vectors with fast similarity search and metadata filters.
Setup	Index shopassist, dimension **768**, metric **cosine**, serverless free tier; created by scripts/setup_pinecone.py.
Used by	Workflows 01 and 02 (insert) and the two retrieval tools in workflow 03.

3.6 PostgreSQL on Supabase

Aspect	Details
What it is	PostgreSQL is an open-source relational (SQL) database; Supabase hosts it for free with a web dashboard.
Tables	orders, tickets (TKT-000N id generated by the database), products, purchase_orders (PO-000N), n8n_chat_histories.
Connection	Session pooler (host aws-…pooler.supabase.com, user postgres.<project>), SSL.
Safety	All queries are **parameterised** ($1, $2…); text from the model is never inserted into SQL directly.

3.7 Slack, Twilio and Gmail SMTP

Aspect	Details
Slack API	Bot posts escalation alerts to #support-escalations (04) and PO approval requests (06).
Twilio SMS	Texts the customer the ticket ID (04) and delay notices (05). Trial accounts send only to verified numbers.
Gmail SMTP	E-mails the full ticket to the support inbox (04), using an app password.
Resilience	All three are set to *continue on error*, so a missing credential or phone number never breaks the chat.

3.8 Docker

Aspect	Details
What it is	Runs applications in isolated containers.
In ShopAssist	n8n runs in Docker Compose; data lives in a volume, so containers can be recreated safely. scripts/fix_webhook_url.sh adds WEBHOOK_URL with an override file.

3.9 Python tooling

Aspect	Details
What it is	Small Python 3 scripts (standard library + pg8000 Postgres driver) that automate setup and testing.
Scripts	build_workflows.py (generates the workflow JSON), import_to_n8n.py (n8n API), setup_database.py, setup_pinecone.py, doctor.py and verify_setup.py (health checks), test_chat.py, chat_cli.py, evaluate.py, kpi_report.py.

Aspect	Details
Why	Repeatable setup ("infrastructure as code"): the whole system can be rebuilt with a few commands, and problems are diagnosed automatically.

3.10 Web app (front end)

Aspect	Details
Back end	app/server.py: Python standard-library HTTP server on port 8501; endpoints /api/chat (forwards to n8n), /api/kpis (database), /api/status (health).
Front end	app/index.html: plain HTML, CSS and JavaScript, no framework. Chat panel, one-tap chips, status lights, KPI cards and live tables refreshed every 10 s.

3.11 Git and GitHub

Aspect	Details
Used for	Version control and submission: all code, workflows, prompts, data, docs and screenshots are in the repository. Secrets stay in .env, which Git ignores.

4. Every n8n node used

Node (type)	Workflows	Purpose in ShopAssist
Manual Trigger	01, 02	Start ingestion by hand
HTTP Request	01, 02	Download the FAQ PDF, urls.txt and each web page
Code (JavaScript)	01–04	Fix MIME type, split URLs, Detect Intent rules, Needs-a-Ticket check, Normalize Ticket
HTML Extract	02	Pull readable body text out of web pages
Basic LLM Chain	02, 04	Summarize pages; build the ticket JSON
Default Data Loader	01, 02	Turn PDF / text into documents with metadata
Recursive Character Text Splitter	01, 02	Chunk 1,000 / overlap 200
Embeddings Ollama	01–03	nomic-embed-text vectors
Pinecone Vector Store	01–03	Insert vectors; retrieve as an agent tool with a metadata filter
Ollama Chat Model	02, 03, 04, 06	qwen2.5:3b for agents, summaries, tickets
Chat Trigger	03	Public hosted chat and chat webhook
AI Agent (Tools Agent)	03, 06	Reasoning and tool calling
Postgres Chat Memory	03	Conversation memory
Postgres Tool	03, 06	order_lookup, stock_lookup, create_purchase_order (agent tools)
Call n8n Workflow Tool	03	escalate_to_human → 04, inventory_expert → 06
Execute Workflow	03	Guaranteed escalation calls 04 directly
Execute Workflow Trigger	04, 06	Entry point of sub-workflows
Structured Output Parser	04	Ticket JSON schema
Postgres	03, 04, 05	Order details, save ticket, update order
IF	03, 05	Ticket missing? / order delayed?
Set (Edit Fields)	02–04, 06	Shape outputs (reply text, ticket ID, metadata)
Slack	04, 06	Escalation alert, PO approval
Twilio	04, 05	SMS to the customer
Send Email (SMTP)	04	E-mail to the support team
Webhook	05	Receive order status updates
Respond to Webhook	05	Return ok / not found to the caller

5. Parameters explained

Parameter	Value	What it controls	Why this value
temperature	0.2 support agent / 0 elsewhere	Randomness of the model	Low = factual and repeatable
numCtx	8192	Context window (tokens the model can read)	Default 2048 cut off the system prompt and tool results
think	false	Reasoning mode on/off	qwen2.5 is not a thinking model; avoids errors
format	json (04)	Force JSON output	Valid ticket JSON
maxIterations	8 (03) / 4 (06)	Max tool-call rounds of an agent	Stops loops; keeps answers fast
contextWindowLength	10	Messages kept in memory	Enough context, small prompt
topK	4	Chunks returned per search	Enough evidence without noise
chunkSize / overlap	1000 / 200	Text chunk length / shared part	One topic per chunk; no lost sentences
dimension / metric	768 / cosine	Vector size / similarity measure	Must match nomic-embed-text
executionTimeout	03: 240 s, 04/06: 150 s, 05: 60 s	Max run time of a workflow	A stuck run cannot block the local model

6. Why a local model? (comparison)

Criterion	**Cloud LLM API (e.g. Gemini, GPT)**	**Local LLM (Ollama + qwen2.5:3b) – chosen**
Cost	Pay per token after free quota	Free
Access	Needs an API key; Gemini returned 403 for the project	No key, no account
Privacy	Customer data leaves the company	Data stays on the laptop
Quality	Higher	Lower – compensated with guardrails
Speed	Fast, independent of laptop	2–23 s per answer on a MacBook Air
Availability	Always on	Only while the laptop runs

7. Evaluation and KPI methods

- **Automatic evaluation** (evaluate.py): 14 questions covering RAG, SQL, combined, missing information, escalation, refusal, one-word inputs and inventory. Each answer is checked for required keywords (and forbidden ones, such as code). Result: 14 / 14, 10 s average.
- **Human review** of the answers, which found wrong details that keywords miss. Those were then fixed with rules.
- **Business KPIs** (kpi_report.py, dashboard): conversations, tickets by type and priority, share resolved without a human (≈ 82 %), low-stock products, purchase orders.
- **Health checks** (doctor.py): code version, .env, Ollama, database tables, Pinecone vector count, n8n workflows and credentials, stuck runs, recent errors.

8. Glossary

Term	Meaning
LLM	Large Language Model – an AI model that reads and writes text
Agent	An LLM that can call tools and act in several steps
Tool	A function the agent can call (database query, search, another workflow)
RAG	Retrieval-Augmented Generation – answer from retrieved documents
Embedding / vector	Numbers that represent the meaning of a text
Vector database	Database that finds the most similar vectors (Pinecone)
Chunk	A small piece of a document stored as one vector
Metadata	Labels stored with a vector (source, url, title)
Hallucination	A confident but invented answer
Guardrail	A rule in code that checks or corrects the AI
Webhook	A URL that another system calls to send data to n8n
Sub-workflow	A workflow called by another workflow
Human-in-the-loop	A person approves or handles what the AI should not decide
Token	A small piece of text (about ¾ of a word) the model processes